

LDAP

Active Directory & LDAP

- (Lightweight Directory Access Protocol) (LDAP)
- In order to enumerate AD, we first need to understand how to communicate with the service. Before we start building our script, let's discuss some theory.
- AD enumeration relies on **LDAP**.
- When a domain machine searches for an object, like a printer, or when we query user or group objects, LDAP is used as the communication channel for the query. In other words, LDAP is the protocol used to communicate with Active Directory.

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- LDAP communication with AD is not always straight-forward, but we'll leverage an Active Directory Services Interface (ADSI) (a set of interfaces built on COM) as an LDAP provider.
- According to [Microsoft's documentation](#), we need a specific LDAP ADsPath in order to communicate with the AD service.
- The LDAP path's prototype looks like this:

```
LDAP://HostName[:PortNumber][/DistinguishedName]
```

• دا الشكل العام لل LDAP

- HOST NAME DOMAIN NAME دا الجزء الخاص ب
- DIS NAME USER /PRINTER /GROUP/ ANY OBJECT الجزء الخاص بال
- PORT هو رقم المنفذ

• ممكن تلاقي مثلا FOREST فيها اكثر من domain ف بيكون طبعا فيهم واحد متحكم وهو دا ال بيتم اضافته لعنوان LDAP

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- In order for LDAP to function, objects in AD (or other directory services) must be formatted according to a specific naming standard. To show an example of a DN, we can use our stephanie domain user.
- We know that stephanie is a user object within the corp.com domain. With this, the DN may look something like this:

CN=Stephanie,CN=Users,DC=corp,DC=com

- الشكل العام للـ dis name لكن م ينفعش تستخدمه كدا

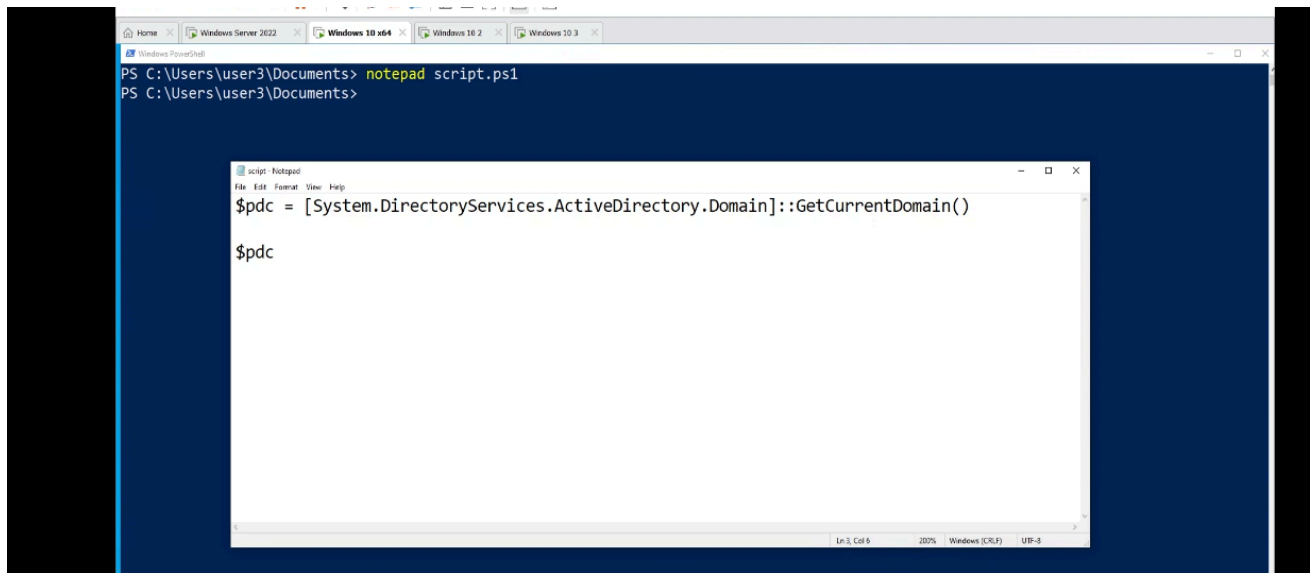
- To enumerate the Primary Domain Controller we use in powershell:

```
[System.DirectoryServices.ActiveDirectory.Domain]::GetCurrentDomain()
```

```
PS C:\Users\victim2\Downloads> [System.DirectoryServices.ActiveDirectory.Domain]::GetCurrentDomain()
```

```
Forest                : green.com
DomainControllers     : {DC1.green.com}
Children              : {}
DomainMode            : Unknown
DomainModeLevel       : 7
Parent                :
PdcRoleOwner          : DC1.green.com
RidRoleOwner          : DC1.green.com
InfrastructureRoleOwner : DC1.green.com
Name                  : green.com
```

- ف بنستعمل الامر دا عشان تقدر تشوف من (pdc) primary role دا ال هو host name



- دا في حاله لو عاوز نشتغل بس script اقدر من حاجه جواه يعني

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- We can use **ADSI** directly in PowerShell to retrieve the DN. We'll use two single quotes to indicate that the search starts at the top of the AD hierarchy.
- `> ([adsis]'').distinguishedName`

```
PS C:\Users\stephanie> ([adsis]'').distinguishedName  
DC=corp,DC=com
```

- his returns the DN in the proper format for the LDAP path.

- دا عشان تعرف ال dis name
- وبكدا نت قدرت تطلع كل حاجه وتقدر تستخدم LDAP

- \$PDC =
[System.DirectoryServices.ActiveDirectory.Domain]::GetCurrentDomain().P
dcRoleOwner.Name
- \$DN = ([adsisearcher]'').distinguishedName
- \$LDAP = "LDAP://\$PDC/\$DN"
- \$direntry = New-Object System.DirectoryServices.DirectoryEntry(\$LDAP)
- \$dirsearcher = New-Object
System.DirectoryServices.DirectorySearcher(\$direntry)\$dirsearcher.FindA
ll()

- دا اسكربت هيطلعلك كل LDAP الموجوده من غير اي مجهود